ECE330 EXAM 1 EQUATION SHEET

sin(x)=cos(x-90°) $\quad \bar{V} = \bar{Z}\bar{I} \quad \bar{S} = \bar{V}\bar{I}^* = P + jQ \quad \bar{S}_{3\varphi} = \sqrt{3}V_L I_L \angle\theta$

$0<\theta<180°$ (lag) $\quad I_L = \sqrt{3}I_\varphi$ (delta) $\quad \bar{Z}_Y = \bar{Z}_\Delta/3$

$-180°<\theta<0$ (lead) $\quad V_L = \sqrt{3}V_\varphi$ (wye) $\quad \mu_0 = 4\pi \times 10^{-7}$H/m

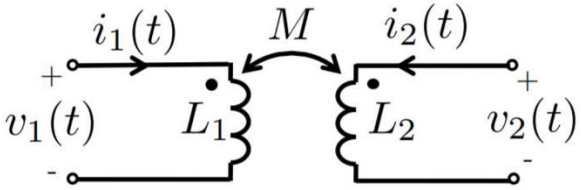
ABC phase sequence has A at 0, B at -120°, and C at +120°

$\int \underline{H} \cdot \underline{dl} = \int \underline{J_f} \cdot \hat{n} dA \quad \int \underline{E} \cdot \underline{dl} = -\frac{d}{dt}\left(\int \underline{B} \cdot \hat{n} dA\right) \quad \mathcal{R} = \frac{l}{\mu A} \quad Ni = \mathcal{R}\varphi$

Fringing $\mathcal{R} = \frac{g}{\mu_0(w+g)(d+g)}$ for width w, depth d, gap g.

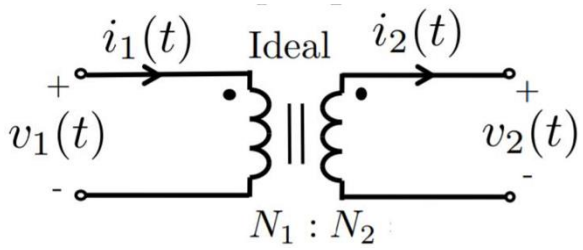
$\varphi = BA \quad \lambda = N\varphi = Li$ (if linear) $\quad v = \frac{d\lambda}{dt} \quad k = \frac{M}{\sqrt{L_1 L_2}}$

1hp=746 W



$v_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$

$v_2 = M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}$



$a = \frac{N_1}{N_2} \quad N_1 i_1 = N_2 i_2$

$\frac{V_1}{V_2} = \frac{N_1}{N_2}$

